MSDM5004
Numerical Methods and Modeling in Science
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Lecture 14

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Chapter 11

Introduction to neural network methods for solving PDEs

Neural network approximation of a function

A neural network (fully connected)

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{W}_N \boldsymbol{\sigma}(\mathbf{W}_{N-1}\boldsymbol{\sigma}(\cdots \boldsymbol{\sigma}(W_2\boldsymbol{\sigma}(W_1\mathbf{x}+\mathbf{b}_1)+\mathbf{b}_2)\cdots+\mathbf{b}_{N-1})+\mathbf{b}_N$$

$\boldsymbol{\theta}$ denotes all the parameters $\sigma(\cdot)$ is the activation function

$\mathbf{x}$: input (vectors)
$\mathbf{y}$: output (vectors)
N: number of hidden layers

w_k: weights (matrix)
b_k: bias (vector)

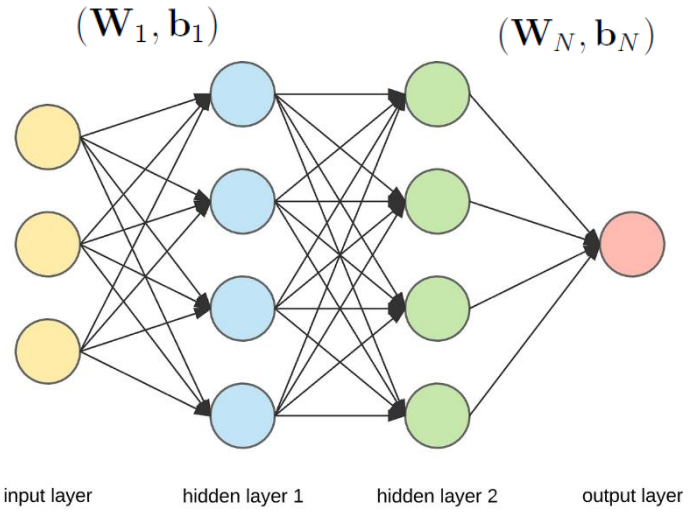
$$\boldsymbol{\sigma}(\mathbf{x}) = \begin{pmatrix} \sigma(x_1) \\ \sigma(x_2) \\ \cdots \\ \sigma(x_d) \end{pmatrix}$$

Two layer neural network

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = W_2\boldsymbol{\sigma}(W_1\mathbf{x}+\mathbf{b}_1)+\mathbf{b}_2$$

Universal approximation theorem [1]

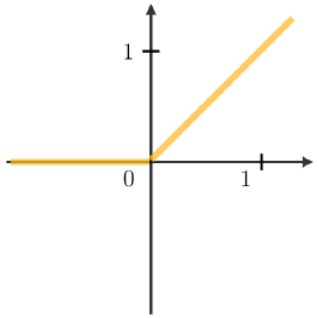
$$f_{\boldsymbol{\theta}}(\mathbf{x}) \approx f(\mathbf{x})$$



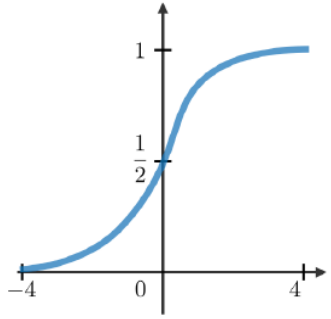
[1] K. Hornik, M. Stinchcombe, and H. White, Multilayer feedforward networks are universal approximators, Neural networks, 2 (1989), pp. 359–366.

Commonly used activation function

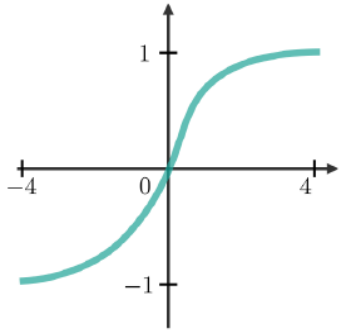
ReLU (Retified Linear Unit): $\sigma(x) = \max\{x, 0\}$



Sigmoid: $\sigma(x) = \dfrac{1}{1+e^{-x}}$



Tanh: $\sigma(x) = \dfrac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$



Supervised learning: trained from labeled data

Determining the neural network parameters $\boldsymbol{\theta}$ by the optimization problem:

Minimize the loss function $\mathcal{L}(\boldsymbol{\theta})$

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{M}\sum_{m=1}^{M} l\left(y_{\boldsymbol{\theta}}^{m}, y_{g}^{m}\right)$$

$\mathbf{y}_g$: ground truth

Example of the loss function

Mean squared error (MSE) $\quad \frac{1}{M}\sum_{m=1}^{M}\left(y_{\boldsymbol{\theta}}^{m} - y_{g}^{m}\right)^{2}$

Training

Solving the optimization problem by gradient descent (GD)

$$\theta_k^{n+1} = \theta_k^n - \frac{\alpha}{M} \sum_{m=1}^{M} \left. \frac{\partial l\left(y_{\boldsymbol{\theta}}^m, y_g^m\right)}{\partial \theta_k} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}^n} \qquad \text{for each } k$$

α: learning rate

Computed by backpropagation (*k* from *N* to *0*) by chain rule formulation

- epoch: one full, complete pass of the entire training dataset through the neural network

Compute partial derivatives by the chain rule

$$\mathbf{y} = w_N \boldsymbol{\sigma}(w_{N-1}\boldsymbol{\sigma}(\cdots(w_2\boldsymbol{\sigma}(w_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2)\cdots + \mathbf{b}_{N-1}) + \mathbf{b}_N)$$

Automatic differentiation

Neural network approximation of *f(x)*

$$f_{\boldsymbol{\theta}}(\mathbf{x}) \approx f(\mathbf{x})$$

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{W}_N \boldsymbol{\sigma}(\mathbf{W}_{N-1}\boldsymbol{\sigma}(\cdots \boldsymbol{\sigma}(W_2\boldsymbol{\sigma}(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) \cdots + \mathbf{b}_{N-1}) + \mathbf{b}_N$$

Automatic differentiation

$$\frac{\partial f}{\partial x_i} \approx \frac{\partial f_{\boldsymbol{\theta}}}{\partial x_i}$$

Also computed by backpropagation (*k* from *N* to *0*) by chain rule formulation

Stochastic gradient decent (SGD)

- For each iteration, a small random subset of data (a mini-batch) is selected.
- The gradient of the loss function is calculated with respect to the model's parameters for only that mini-batch.

[2] L. Bottou, Large-Scale Machine Learning with Stochastic Gradient Descent, Proceedings of COMPSTAT 2010, Paris, 22-27 August 2010, 177-186.

Adaptive moment estimation (Adam)

- Computes individual adaptive learning rates for different parameters from estimates of first and second moments of the gradients.

[3] D. P. Kingma, J. Ba, Adam: A Method for Stochastic Optimization, ICLR 2015.

Physically informed neural networks (PINN)

A PDE, e.g.

$$u_t + \mathcal{N}[u] = f(\mathbf{x}, t), \ \mathbf{x} \in \Omega, \ t \in [0, T]$$

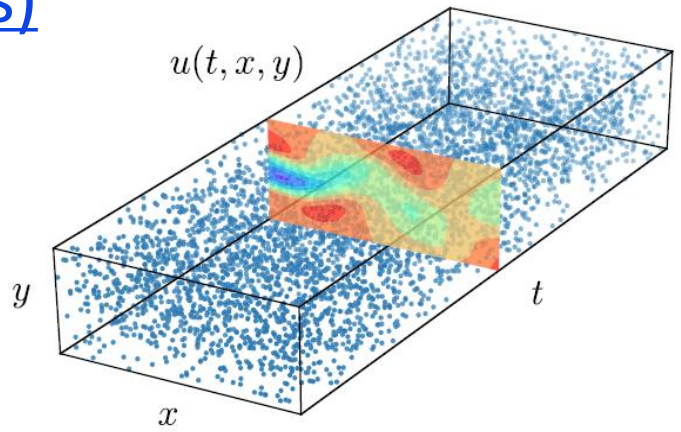
where $\mathcal{N}[\cdot]$ is a differential operator in space.

For diffusion equation in two dimensions, $\mathcal{N}[u] = -u_{xx} - u_{yy}, \ f(\mathbf{x}, t) = 0.$

Sampling points (collocation points)

$$(\mathbf{x}_i, t_i), \ i = 1, 2, \cdots, N$$

can be random



[4] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics, 378, 686–707, 2019.

[5] GE Karniadakis, IG Kevrekidis, L Lu, P Perdikaris, S Wang, L Yang, Physics-informed machine learning, Nature Reviews Physics 3 (6), 422-440, 2021.

Loss

Initial-boundary value problem of the PDE

$$u_t + \mathcal{N}[u] = f(\mathbf{x}, t), \ \mathbf{x} \in \Omega, \ t \in [0, T]$$

$$u = g, \ \mathbf{x} \in \partial\Omega$$

$$u = u_0, \ t = 0$$

$\{(\mathbf{x}_f^i, t_f^i)\}$: points inside the domain
$\{(\mathbf{x}_b^i, t_b^i)\}$: points on $\partial\Omega$
$\{(\mathbf{x}_0^i, 0)\}$: points at $t = 0$

Mean squared error

$$\mathrm{MSE} = \mathrm{MSE}_f + \mathrm{MSE}_0 + \mathrm{MSE}_b$$

$$\mathrm{MSE}_f = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| u_t(\mathbf{x}_f^i, t_f^i) + \mathcal{N}[u](\mathbf{x}_f^i, t_f^i) - f(\mathbf{x}_f^i, t_f^i) \right|^2$$

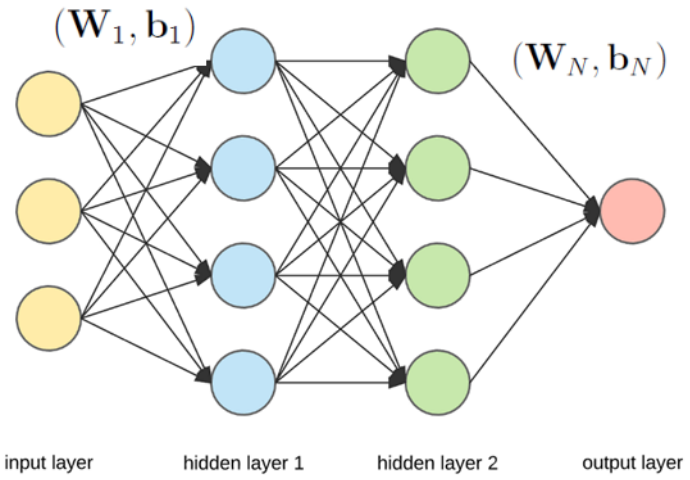
$$\mathrm{MSE}_b = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| u(\mathbf{x}_b^i, t_b^i) - g(\mathbf{x}_b^i, t_b^i) \right|^2$$

$$\mathrm{MSE}_0 = \frac{1}{N_0} \sum_{i=1}^{N_0} \left| u(\mathbf{x}_0^i, 0) - u_0(\mathbf{x}_0^i) \right|^2$$

unsupervised learning
(not based on data)

Neural network structure

Fully-connected neural network



2-5 layers

Example

Schrödinger equation

$$ih_t + 0.5h_{xx} + |h|^2 h = 0, \quad x \in [-5, 5], \quad t \in [0, \pi/2],$$

$$h(0, x) = 2\,\mathrm{sech}(x),$$

$$h(t, -5) = h(t, 5),$$

$$h_x(t, -5) = h_x(t, 5),$$

$N_f = 20000$ points, $N_b = 50$ points, $N_0 = 50$ points, random

$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| ih_t(t_f^i, x_f^i) + 0.5h_{xx}(t_f^i, x_f^i) + |h(t_f^i, x_f^i)|^2 h(t_f^i, x_f^i) \right|^2$$

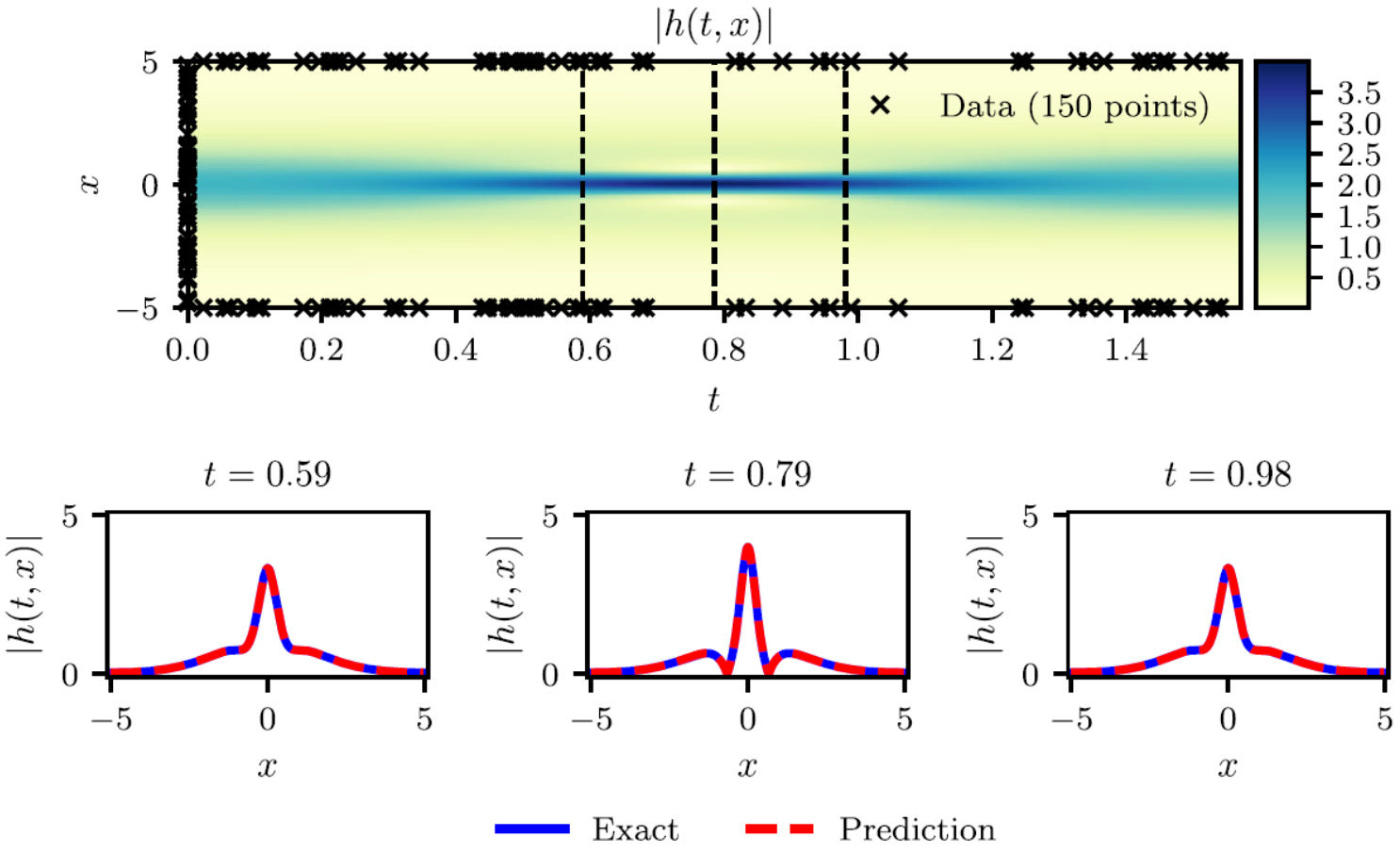
$$MSE_b = \frac{1}{N_b} \sum_{i=1}^{N_b} \left(|h^i(t_b^i, -5) - h^i(t_b^i, 5)|^2 + |h_x^i(t_b^i, -5) - h_x^i(t_b^i, 5)|^2 \right)$$

$$MSE_0 = \frac{1}{N_0} \sum_{i=1}^{N_0} |h(0, x_0^i) - h_0^i|^2$$

$$h(t, x) = \begin{bmatrix} u(t, x) & v(t, x) \end{bmatrix}$$

- 5-layer deep neural network with 100 neurons per layer
- a hyperbolic tangent activation function

Results



Inverse problem

Navier–Stokes equation

$$u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy}),$$
$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}),$$

where $u(t, x, y)$ denotes the x-component of the velocity field, $v(t, x, y)$ the y-component, and $p(t, x, y)$ the pressure.

$\lambda = (\lambda_1, \lambda_2)$ are the unknown parameters

Assume $u = \psi_y, \quad v = -\psi_x$

to satisfy the incompressible condition $u_x + v_y = 0$

approximating $[\psi(t, x, y) \quad p(t, x, y)]$ using a single neural network with two outputs.

Inverse problem:

Given noisy measurements $\{t^i, x^i, y^i, u^i, v^i\}_{i=1}^N$ to determine $\lambda = (\lambda_1, \lambda_2)$

Denote

$$f := u_t + \lambda_1(uu_x + vu_y) + p_x - \lambda_2(u_{xx} + u_{yy}),$$
$$g := v_t + \lambda_1(uv_x + vv_y) + p_y - \lambda_2(v_{xx} + v_{yy}),$$

The mean squared error is

$$MSE := \frac{1}{N}\sum_{i=1}^{N}\left(|u(t^i, x^i, y^i) - u^i|^2 + |v(t^i, x^i, y^i) - v^i|^2\right) \cdot$$

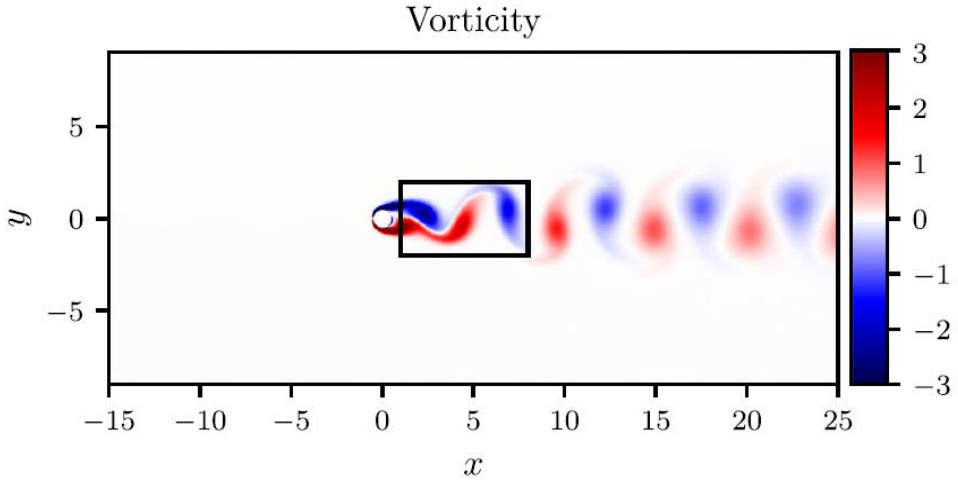
$$+ \frac{1}{N}\sum_{i=1}^{N}\left(|f(t^i, x^i, y^i)|^2 + |g(t^i, x^i, y^i)|^2\right)$$

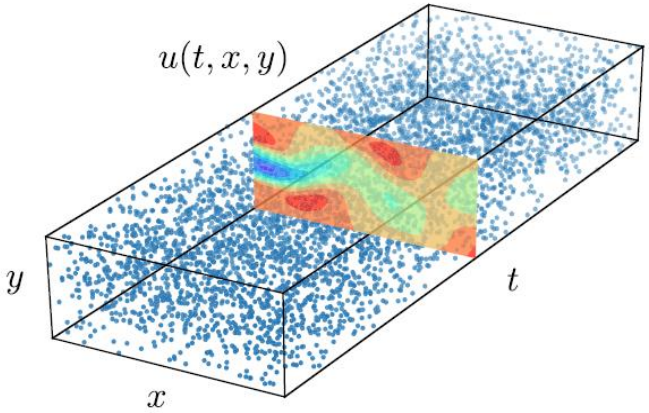
Train the neural network to find $\begin{bmatrix}\psi(t,x,y) & p(t,x,y)\end{bmatrix}$

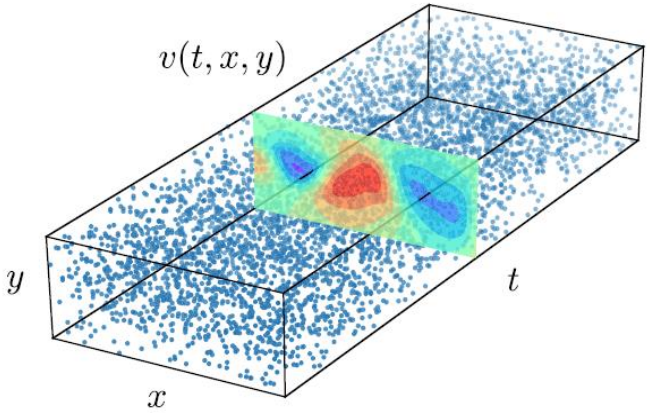
and $\lambda = (\lambda_1, \lambda_2)$

9 layers with 20 neurons in each layer

Results:







$N = 5{,}000$

Correct PDE	$u_t + (uu_x + vu_y) = -p_x + 0.01(u_{xx} + u_{yy})$ $v_t + (uv_x + vv_y) = -p_y + 0.01(v_{xx} + v_{yy})$
Identified PDE (clean data)	$u_t + 0.999(uu_x + vu_y) = -p_x + 0.01047(u_{xx} + u_{yy})$ $v_t + 0.999(uv_x + vv_y) = -p_y + 0.01047(v_{xx} + v_{yy})$
Identified PDE (1% noise)	$u_t + 0.998(uu_x + vu_y) = -p_x + 0.01057(u_{xx} + u_{yy})$ $v_t + 0.998(uv_x + vv_y) = -p_y + 0.01057(v_{xx} + v_{yy})$

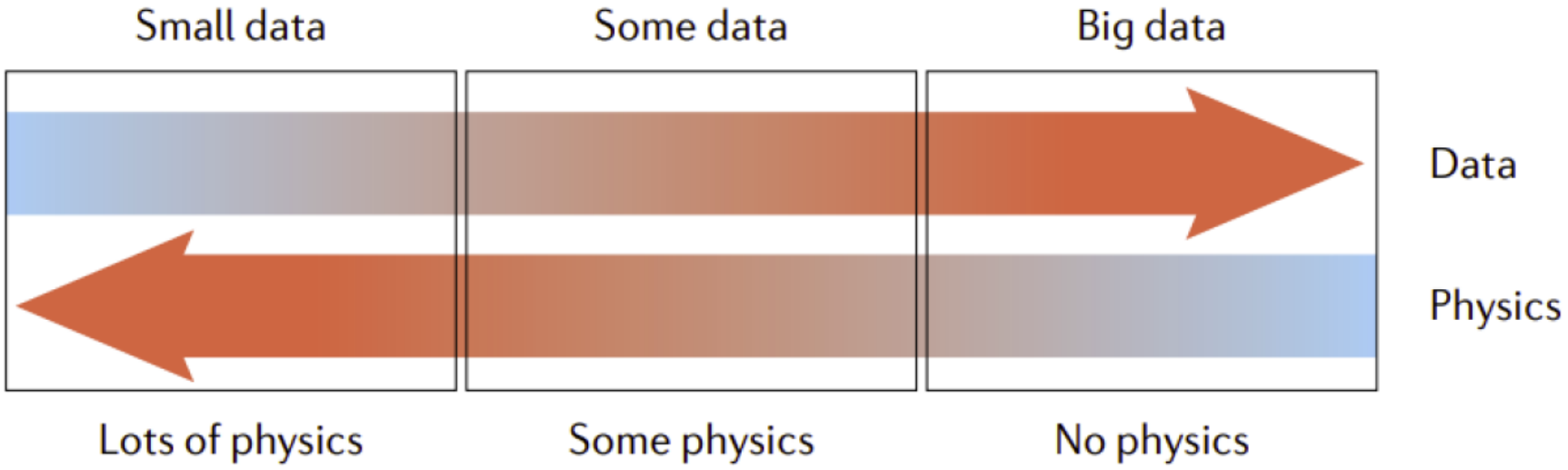
Neural network/numerical solutions of PDEs

- In low dimensions ($d \leq 3$)
  - Traditional numerical methods typically work well
  - Deep learning methods: high coding efficiency

    suitable for inverse problems

- In high dimensions
  - Tradition methods fail (Curse of dimensionality)
  - Deep learning methods work

Three possible categories of physical problems and associated available data



- Small data regime: we know all the physics and coefficients of PDE, as well as data for initial and boundary conditions.

- The middle one: we know some data and some physics, possibly miss some parameters or even an entire term in PDE.

- Big data regime: we may not know any of the physics, but we have a big amount of data. Then a data-driven approach may be most effective. For example, using operator regression methods to discover new physics.